

# ASP.NET Core MVC Architecture

## Directory Layout & Component Breakdown

An ASP.NET Core MVC application enforces a strict separation of concerns by isolating user interfaces, request handling, and business data into distinct components. Below is the standard folder structure for a monolithic ASP.NET Core MVC project along with an explanation of each directory.

## Standard ASP.NET Core MVC Directory Structure

```
MyMvcApp/
├── Controllers/
│   ├── HomeController.cs
│   └── AccountController.cs
├── Models/
│   ├── ErrorViewModel.cs
│   └── UserViewModel.cs
├── Views/
│   ├── Home/
│   │   ├── Index.cshtml
│   │   └── Privacy.cshtml
│   ├── Shared/
│   │   ├── _Layout.cshtml
│   │   ├── _ValidationScriptsPartial.cshtml
│   │   └── Error.cshtml
│   ├── _ViewImports.cshtml
│   └── _ViewStart.cshtml
├── wwwroot/
│   ├── css/
│   ├── js/
│   └── lib/
├── Data/ (or Infrastructure/)
│   └── ApplicationDbContext.cs
├── Services/
│   ├── IUserService.cs
│   └── UserService.cs
├── appsettings.json
└── Program.cs
```

## Folder & File Breakdown

- **Controllers/**
  - Contains C# classes derived from `Controller` or `ControllerBase`.
  - Responsible for intercepting HTTP requests, calling business services, populating data models, and returning an appropriate View response (e.g., `return View(model);`).
- **Models/**
  - Contains C# classes representing the data passed between Controllers and Views (ViewModels or Data Transfer Objects).
  - Houses data validation rules (such as `[Required]` or `[StringLength]`) used by the framework to automatically validate form inputs.
- **Views/**
  - Houses Razor template files ( `.cshtml` ) that mix HTML markup with C# syntax to render output dynamically.

- **Feature-Specific Folders (Views/Home/, Views/Account/):** View files are grouped in folders named after their corresponding Controller (e.g., actions inside `HomeController.cs` render views from `Views/Home/` ).
- **Views/Shared/:** Stores shared layout pages and partial views accessible across the entire application:
  - `_Layout.cshtml` : The master UI shell (header, navigation, footer, global CSS/JS imports).
  - `Error.cshtml` : Default view rendered when uncaught exceptions occur.
- `_ViewStart.cshtml` : Executes before any individual view renders, typically defining the default shared layout ( `Layout = "_Layout";` ).
- `_ViewImports.cshtml` : Declares global Razor directives (such as `@using MyMvcApp.Models` or `@addTagHelper` ) so individual views do not need to re-import common namespaces.
- **wwwroot/**
  - The web-root directory containing static assets served directly to the browser (CSS files, client-side JavaScript, images, third-party libraries like Bootstrap or jQuery).
- **Data/ or Services/**
  - Houses database contexts (Entity Framework `DbContext` ), repository classes, and core domain/business logic interfaces and implementations to keep controllers thin.
- **Program.cs & appsettings.json**
  - **Program.cs:** The entry point of the application where services are registered in the Dependency Injection (DI) container and the HTTP request pipeline middleware (routing, authentication, static files) is configured.
  - **appsettings.json:** Stores application settings, connection strings, and environment-specific configuration keys.